

```

PS C:\Users\marzh> $lab = "C:\Users\marzh\Desktop\omp_labs"
PS C:\Users\marzh> New-Item -ItemType Directory -Force -Path $lab | Out-Null
PS C:\Users\marzh> $zip = Get-ChildItem -Path "$HOME\Downloads" -Filter "omp_labs*.zip"
| Sort-Object LastWriteTime -Descending | Select-Object -First 1
PS C:\Users\marzh> $zip.FullName
C:\Users\marzh\Downloads\omp_labs.zip
PS C:\Users\marzh> $tmp = Join-Path $env:TEMP "omp_x"
PS C:\Users\marzh> Remove-Item $tmp -Recurse -Force -ErrorAction SilentlyContinue
PS C:\Users\marzh> Expand-Archive -Path $zip.FullName -DestinationPath $tmp -Force
PS C:\Users\marzh> Get-ChildItem "$tmp\omp_labs" -Force | Copy-Item -Destination $lab -
Recurse -Force
PS C:\Users\marzh> Set-Location $lab
PS C:\Users\marzh\Desktop\omp_labs> dir

```

Каталог: C:\Users\marzh\Desktop\omp_labs

Mode	LastWriteTime		Length	Name
----	-----		-----	----
d-----	25.09.2026	13:52		lab1
d-----	25.09.2026	13:52		lab2
d-----	25.09.2026	13:52		lab3
d-----	25.09.2026	13:52		lab4
d-----	25.09.2026	13:52		lab5
d-----	25.09.2026	13:52		lab6
-a----	25.09.2026	13:50	49	.gitignore
-a----	25.09.2026	13:50	11298	make_plots.py
-a----	25.09.2026	13:50	8005	README.md
-a----	25.09.2026	13:50	3456	run_all.bat
-a----	25.09.2026	13:50	6908	run_all.ps1

```

PS C:\Users\marzh\Desktop\omp_labs> Set-ExecutionPolicy -Scope Process -ExecutionPolicy
Bypass
PS C:\Users\marzh\Desktop\omp_labs> .\run_all.ps1

```

```
=====
OpenMP Lab Practice Manual - Java Edition - Full Suite
=====
```

```
=====
OPENMP LAB PRACTICE MANUAL - JAVA EDITION
Student: Marzhan Sherekhan (230109015)
=====
```

```
Timestamp : 2026-09-25 13:52:53
Machine   : LAPTOP
OS        : Майкрософт Windows 11 Домашняя для одного языка
CPU       : 12th Gen Intel(R) Core(TM) i7-12700H
Cores     : 14 physical / 20 logical
Java      : java version "17.0.10" 2024-01-16 LTS
=====
```

```
Capturing hardware specification...
Saved results\hw_info.txt
```

```
Compiling all six labs...
```

```
lab1\Lab1ForkJoin.java:40: error: cannot find symbol
    long osTid = Thread.currentThread().threadId();
                                   ^
```

```
symbol:   method threadId()
location: class Thread
```

```
1 error
```

```
COMPILATION FAILED.
```

```
PS C:\Users\marzh\Desktop\omp_labs> python make_plots.py
```

```
skip: lab1_oversubscription.csv not found
```

```
skip: lab2_scaling.csv not found
```

```
skip: lab3_sweep.csv not found
```

```
skip: lab3_imbalance.csv not found
```

```
skip: lab4_scaling.csv not found
```

```
skip: lab5_cutoff.csv not found
```

```
skip: lab6_pipeline.csv not found
```

```
Generated:
```

```
nothing - run the labs first so the CSVs exist
```

```
PS C:\Users\marzh\Desktop\omp_labs> (Get-Content lab1\Lab1ForkJoin.java) -replace '\.threadId\(\)\', '.getId()' | Set-Content lab1\Lab1ForkJoin.java
```

```
PS C:\Users\marzh\Desktop\omp_labs> Select-String -Path lab1\Lab1ForkJoin.java -Pattern "getId\(\)"
```

```
lab1\Lab1ForkJoin.java:40:                long osTid = Thread.currentThread().getId();
```

```
PS C:\Users\marzh\Desktop\omp_labs> .\run_all.ps1
```

```
=====
OpenMP Lab Practice Manual - Java Edition - Full Suite
=====
```

```
=====
OPENMP LAB PRACTICE MANUAL - JAVA EDITION
Student: Marzhan Sherekhan (230109015)
=====
```

```
Timestamp : 2026-09-25 13:54:51
Machine   : LAPTOP
OS        : Майкрософт Windows 11 Домашняя для одного языка
CPU       : 12th Gen Intel(R) Core(TM) i7-12700H
Cores     : 14 physical / 20 logical
Java      : java version "17.0.10" 2024-01-16 LTS
=====
```

```
Capturing hardware specification...
  Saved results\hw_info.txt
```

```
Compiling all six labs...
  Compiled OK.
```

```
Running LAB 1 - Fork-Join Model & Thread Teams ...
```

```
#####
# LAB 1 - Fork-Join Model & Thread Teams
# command: java -cp out Lab1ForkJoin
# started: 13:54:53
#####
```

```
=====
LAB 1 - FORK-JOIN MODEL, TEAM CREATION AND THREAD SCOPING
=====
```

```
Available processors : 20
JVM                  : Java HotSpot(TM) 64-Bit Server VM 17.0.10
Common pool parallelism: 19
```

```
TASK 1.1 - VERIFICATION OF NON-DETERMINISM (10 consecutive runs)
=====
```

```
run 1 completion order: 2 3 1 0
run 2 completion order: 2 3 0 1
run 3 completion order: 2 3 0 1
run 4 completion order: 2 3 0 1
run 5 completion order: 2 1 3 0
run 6 completion order: 2 3 1 0
run 7 completion order: 2 3 1 0
run 8 completion order: 2 3 0 1
run 9 completion order: 2 3 1 0
run 10 completion order: 2 3 1 0
```

```
Runs matching run 1's order: 4 of 9 -> NON-DETERMINISTIC (expected)
```

Saved lab1_nondeterminism.txt

TASK 1.2 – THREAD OVERSUBSCRIPTION SWEEP (team lifecycle cost)

P	run2 (ms)	run3 (ms)	mean (ms)
1	0.2605	0.1532	0.2069
2	0.5700	0.4205	0.4952
4	2.0776	0.3886	1.2331
8	0.7695	0.6767	0.7231
16	1.2729	0.9319	1.1024
32	1.5684	1.0764	1.3224
64	2.6378	1.5667	2.1022

Saved lab1_oversubscription.csv

TASK 1.3 – CPU CORE SATURATION ANALYSIS

Each thread computes 10,000,000 square roots.

Watch Task Manager > Performance > CPU while this runs.

P	mean (ms)	total sqrt ops	ops/sec	vs P=1
1	38.13	10,000,000	262,275,476	1.000x
2	21.89	20,000,000	913,477,678	0.574x
4	23.26	40,000,000	1,719,742,211	0.610x
8	43.92	80,000,000	1,821,292,503	1.152x
16	55.32	160,000,000	2,892,488,028	1.451x
32	90.40	320,000,000	3,539,633,106	2.371x
64	142.51	640,000,000	4,490,840,440	3.738x

Saved lab1_saturation.csv

LAB 1 COMPLETE

done.

Running LAB 2 – Numerical Integration & Reductions ...

```
#####
# LAB 2 – Numerical Integration & Reductions
# command: java -cp out Lab2PiIntegration
# started: 13:54:55
#####
```

LAB 2 – NUMERICAL INTEGRATION (PI) AND PARALLEL REDUCTIONS

```
Available processors : 20
Integration steps N  : 100,000,000
True Pi              : 3.141592653589793
```

Warming up JIT...

Warm-up complete.

TASK 2.1 - RACE CONDITION QUANTIFICATION (Variant A)

N = 1,000,000 steps, threads P in {1,2,4,8}

P	Computed Pi	Abs Error	Time (ms)
1	3.141592653590	2.886580e-14	8.55
2	1.854590436003	1.287002e+00	3.23
4	0.664965067440	2.476628e+00	2.47
8	0.833368501806	2.308224e+00	2.11

Saved lab2_race.csv

TASK 2.2 - CRITICAL SECTION OVERHEAD (Variant B)

N = 1,000,000 steps

Single-threaded baseline	: Pi = 3.141592653590		8.42 ms	
synchronized, P = 1	: Pi = 3.141592653590		16.91 ms	2.01x slower c
contention 100.8%				
synchronized, P = 2	: Pi = 3.141592653590		8.35 ms	0.99x slower c
contention -0.9%				
synchronized, P = 4	: Pi = 3.141592653590		22.33 ms	2.65x slower c
contention 165.1%				
synchronized, P = 8	: Pi = 3.141592653590		19.50 ms	2.32x slower c
contention 131.5%				

Saved lab2_critical.csv

TASK 2.3 & 2.4 - STRONG SCALING OF VARIANT C (5 trials each)

N = 100,000,000 steps, P in {1,2,4,8,16}

P	Mean T(P) ms	Computed Pi	Abs Error	S(P)	E(P)
1	751.27	3.141592653590	4.441e-16	1.000x	100.0%
2	382.30	3.141592653590	4.441e-16	1.965x	98.3%
4	208.15	3.141592653590	4.441e-16	3.609x	90.2%
8	128.10	3.141592653590	4.441e-16	5.865x	73.3%
16	88.34	3.141592653590	4.441e-16	8.504x	53.2%

Saved lab2_scaling.csv

LAB 2 COMPLETE

done.

Running LAB 3 - Work-Sharing & Loop Scheduling ...

```
#####
# LAB 3 - Work-Sharing & Loop Scheduling
# command: java -cp out Lab3Mandelbrot
# started: 13:55:03
#####
```

```

=====
LAB 3 - WORK-SHARING AND LOOP SCHEDULING POLICIES (MANDELBROT)
=====
Available processors : 20
Image                : 1920 x 1080, MAX_ITER = 1000
Sweep                : P in {2,4,8,16} x C in {1,16,64,256}, 3 runs each

Warming up JIT...
Warm-up complete.

TASK 3.1 & 3.4 - STATIC SCHEDULING BASELINE AND LOAD IMBALANCE
-----
  Threads |      Mean (ms) | Total iterations |      Imbalance | per-thread work
-----
  2       |      277.91 |    203,116,879 |      0.0107 | max=102,099,479 min=101,0
17,400
  4       |      273.99 |    203,116,879 |      1.9736 | max=101,156,033 min=938,4
97
  8       |      227.39 |    203,116,879 |      3.2622 | max=83,203,218 min=376,03
7
  16      |      156.55 |    203,116,879 |      4.0220 | max=51,209,114 min=151,21
0

GUIDED SCHEDULING (for comparison)
-----
  2       |      275.08 |    203,116,879 |      0.1732 | max=110,354,310 min=92,76
2,569
  4       |      151.41 |    203,116,879 |      0.1811 | max=53,811,611 min=44,616
,609
  8       |      85.87  |    203,116,879 |      0.3122 | max=28,645,575 min=20,719
,110
  16      |      73.88  |    203,116,879 |      0.4121 | max=14,597,781 min=9,366,
710
  Saved lab3_imbalance.csv

TASK 3.2 - 4x4 PARAMETER SWEEP (dynamic scheduling)
-----
  P \ C   |      C=1 |      C=16 |      C=64 |      C=256 |
-----
  2       | 252.48 ms | 255.50 ms | 261.98 ms | 305.76 ms |
  4       | 137.25 ms | 139.48 ms | 144.22 ms | 305.90 ms |
  8       | 82.01 ms  | 84.28 ms  | 140.43 ms | 318.66 ms |
  16      | 74.87 ms  | 63.56 ms  | 159.88 ms | 307.24 ms |

  Saved lab3_sweep.csv

=====
LAB 3 COMPLETE
=====
done.

```

Running LAB 4 - Cache Coherency & False Sharing ...

```
#####
# LAB 4 - Cache Coherency & False Sharing
# command: java -cp out Lab4FalseSharing
# started: 13:55:18
#####
```

```
=====
=
LAB 4 - MEMORY HIERARCHY, CACHE COHERENCY AND FALSE SHARING
=====
=
```

```
Available processors : 20
Increments per thread: 100,000,000
Padding stride       : 8 longs = 64 bytes (one 64-byte line)
```

```
Warming up JIT...
Warm-up complete.
```

TASKS 4.1, 4.2, 4.3 - SCALING PROFILE OF ALL THREE VARIANTS

-						
P	Unpadded (ms)	Padded (ms)	ThreadLocal (ms)	FS penalty	TL speed	
up						
1	182.65	171.76	1.57	1.063x	116.48	
5x						
2	370.33	683.37	1.61	0.542x	229.70	
5x						
4	2051.91	540.57	3.18	3.796x	644.31	
4x						
8	3052.27	645.27	3.49	4.730x	875.46	
5x						
16	6288.93	696.48	10.07	9.030x	624.74	
4x						

Saved lab4_scaling.csv

Cache-line arithmetic:

sizeof(long) = 8 bytes, cache line = 64 bytes -> 8 counters share one line
unpadded at P=16 : 16 x 8 = 128 bytes -> spans 2 cache lines
padded at P=16 : stride 64 bytes -> 16 distinct lines

```
=====
=
LAB 4 COMPLETE
=====
=
```

Running LAB 5 - Task-Based Parallel Merge Sort ...

```
#####
# LAB 5 - Task-Based Parallel Merge Sort
# command: java -cp out Lab5MergeSort
# started: 13:56:02
#####
```

LAB 5 - RECURSIVE TASK-BASED PARALLELISM (PARALLEL MERGE SORT)

```
=====
Available processors : 20
Array size          : 5,000,000 random integers
Cutoff sweep K      : [1, 10, 100, 1000, 10000, 50000, 100000]
```

TASK 5.1 - ALGORITHM VERIFICATION

```
-----
Output array sorted ascending : PASS
first 5  : [0, 0, 0, 1, 1]
last 5   : [9999989, 9999989, 9999990, 9999991, 9999992]
```

WORK (T_1) - SEQUENTIAL BASELINE

```
-----
Arrays.sort single-threaded T_1 = 331.71 ms
```

TASK 5.2 - CUTOFF THRESHOLD OPTIMISATION SWEEP

```
-----
```

Cutoff K	Mean (ms)	Tasks created	Speedup	Sorted
1	265.87	5,805,695	1.248x	PASS
10	168.86	1,048,575	1.964x	PASS
100	150.11	131,071	2.210x	PASS
1000	102.15	16,383	3.247x	PASS
10000	131.18	1,023	2.529x	PASS
50000	120.49	255	2.753x	PASS
100000	120.99	127	2.742x	PASS

```
-----
```

```
Optimal cutoff K = 1,000 at 102.15 ms (3.247x over sequential)
Saved lab5_cutoff.csv
```

TASK 5.3 - WORK-SPAN (CRITICAL PATH) MODEL

```
-----
```

Work T_1 (Arrays.sort, 1 core)	:	331.71 ms
T_1 of the task algorithm (pool size 1)	:	425.70 ms
T_P (best cutoff, full pool)	:	102.15 ms
Recursion levels above cutoff $\log_2(N/K)$	:	12.29
Measured parallelism $T_1(\text{task}) / T_P$	:	4.167
Available hardware parallelism	:	20
Parallel efficiency achieved	:	20.8%

```
-----
Saved lab5_workspan.csv
```

```

=====
done.
Running LAB 6 - Pipeline Parallelism (BONUS) ...

#####
# LAB 6 - Pipeline Parallelism (BONUS)
# command: java -cp out Lab6Pipeline
# started: 13:56:09
#####

=====
=====
LAB 6 (BONUS) - PIPELINE PARALLELISM AND WAVEFRONT PROCESSING
=====
=====
Available processors : 20
Packets             : 3,000 tiles of 64x64
Queue capacity       : 32 (bounded, backpressure enabled)

Warming up JIT...
Warm-up complete.

THROUGHPUT AND BOTTLENECK ANALYSIS ACROSS FILTER WORKLOADS
-----
-----
Intensity | Total (ms) | Pkts/sec | Latency ms | avg Q1 | avg Q2 | Bottle
neck
-----
-----
1 | 72.30 | 41493.6 | 0.0455 | 0.36 | 0.36 | Stage
1 (Producer) - I/O-BOUND
4 | 210.91 | 14223.9 | 2.3633 | 31.08 | 0.08 | Stage
2 (Filter) - COMPUTE-BOUND
16 | 750.79 | 3995.8 | 8.4364 | 31.73 | 0.06 | Stage
2 (Filter) - COMPUTE-BOUND
64 | 2845.77 | 1054.2 | 32.0613 | 31.81 | 0.02 | Stage
2 (Filter) - COMPUTE-BOUND

Stage busy-time breakdown (identifies the rate limiter):
intensity 1 : producer 41.1% filter 50.3% consumer 8.7%
intensity 4 : producer 15.5% filter 80.6% consumer 3.9%
intensity 16 : producer 4.8% filter 93.8% consumer 1.4%
intensity 64 : producer 1.4% filter 97.9% consumer 0.6%

Saved lab6_pipeline.csv

=====
=====
LAB 6 COMPLETE
=====
=====

```

```
intensity 4 : producer 15.5% filter 80.6% consumer 3.9%
intensity 16 : producer 4.8% filter 93.8% consumer 1.4%
intensity 64 : producer 1.4% filter 97.9% consumer 0.6%
```

Saved lab6_pipeline.csv

```
=====
=====
LAB 6 COMPLETE
=====
=====
done.
```

```
=====
SUITE COMPLETE - 2026-09-25 13:56:17
=====
```

All six labs complete.

Transcript : C:\Users\marzh\Desktop\omp_labs\results\suite_output.txt

CSV data : C:\Users\marzh\Desktop\omp_labs\results

Name	Length
----	-----
hw_info.txt	1694
lab1_nondeterminism.txt	3557
lab1_oversubscription.csv	392
lab1_saturation.csv	547
lab2_critical.csv	443
lab2_race.csv	256
lab2_scaling.csv	654
lab3_imbalance.csv	666
lab3_sweep.csv	1387
lab4_scaling.csv	1101
lab5_cutoff.csv	565
lab5_workspan.csv	448
lab6_pipeline.csv	637
suite_output.txt	32174

PS C:\Users\marzh\Desktop\omp_labs> python make_plots.py

Generated:

```
results/lab1_oversubscription.png
results/speedup_efficiency.png
results/lab3_heatmap.png
results/lab3_imbalance.png
results/lab4_false_sharing.png
results/lab5_cutoff.png
results/lab6_pipeline.png
```

PS C:\Users\marzh\Desktop\omp_labs> |